

RAMNICU SARAT, Romania

WMO: 153070

Lat: 45.3906N

Lon: 27.0386E

Elev: 154

StdP: 99.49

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.4	-8.6	-13.7	1.2	-8.4	-11.8	1.4	-6.7	11.9	-1.3	9.4	-2.7	2.1	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.6	21.6	31.2	21.0	29.9	20.6	23.6	29.6	22.6	28.7	21.7	27.9	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	16.9	26.7	20.6	15.6	25.3	19.6	14.6	24.5	71.2	29.8	67.1	28.5	63.9	28.1	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	6.7	5.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-1.0	2.4	6.2	11.3	17.2	21.6	22.7	23.0	17.9	11.6	6.1	1.5		
	DBStd	9.24	4.26	4.71	4.78	4.31	3.31	2.91	2.76	3.21	4.00	4.30	4.59	4.42		
	HDD10.0	1158	340	215	133	37	1	0	0	0	0	33	133	265		
	HDD18.3	2862	598	446	375	214	59	7	2	4	57	212	366	523		
	CDD10.0	1803	0	3	17	75	225	349	394	404	237	82	17	1		
	CDD18.3	466	0	0	0	2	25	106	138	150	43	2	0	0		
	CDH23.3	4048	0	0	2	26	213	868	1201	1367	356	17	0	0		
	CDH26.7	1389	0	0	0	3	40	263	432	546	104	1	0	0		
(14) Wind	WSAvg	2.3	2.1	2.3	2.7	2.6	2.4	2.5	2.3	2.4	2.2	2.1	1.9	2.2		
(15) Precipitation	PrecAvg	538	27	23	31	43	66	72	62	47	50	49	32	35		
	PrecMax	786	68	62	68	105	172	132	152	117	104	163	97	95		
	PrecMin	365	1	3	1	20	12	11	4	7	10	3	0	1		
	PrecStd	116	19	19	18	21	41	29	36	30	29	35	25	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	17.7	22.5	26.1	31.0	32.6	34.5	34.9	32.1	25.9	20.3	13.7		
		MCWB	8.2	11.9	13.7	16.4	21.0	22.5	24.2	22.0	19.5	18.0	13.8	9.5		
	2%	DB	9.0	14.4	19.5	23.2	27.8	30.9	32.4	32.8	29.2	22.9	17.4	11.4		
		MCWB	6.0	10.4	12.0	15.0	18.6	21.8	22.5	20.8	18.3	16.4	12.3	8.0		
	5%	DB	6.5	11.9	16.4	20.9	25.8	29.5	30.7	31.4	27.1	20.6	14.7	9.3		
		MCWB	4.3	8.2	10.3	13.6	17.7	21.1	21.4	20.5	17.7	14.8	11.1	6.9		
	10%	DB	4.7	9.5	13.8	18.9	24.1	28.0	29.2	29.8	24.8	18.4	12.5	7.3		
		MCWB	3.3	6.8	8.9	12.7	16.9	20.5	20.7	20.2	17.1	13.7	9.9	5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	12.9	15.4	17.1	21.5	24.4	25.9	24.2	21.3	19.5	14.5	10.6		
		MCDB	11.5	16.9	21.3	25.3	27.5	29.4	30.9	31.3	28.1	23.9	19.7	12.3		
	2%	WB	6.2	10.5	12.3	15.6	19.7	23.0	24.1	22.9	19.7	17.3	13.0	8.9		
		MCDB	8.2	14.0	18.0	22.3	25.6	29.3	30.1	29.9	25.3	21.0	16.4	11.1		
	5%	WB	4.7	8.6	10.7	14.4	18.7	22.1	22.7	21.9	18.8	15.7	11.7	7.3		
		MCDB	6.2	11.5	15.7	20.1	24.2	28.3	28.0	29.2	25.5	19.3	14.3	9.5		
	10%	WB	3.3	7.0	9.3	13.2	17.8	21.1	21.7	21.1	17.8	14.5	10.3	5.6		
		MCDB	4.5	9.1	13.1	18.1	22.7	27.3	27.2	27.7	24.1	17.7	12.3	7.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.7	8.9	9.8	10.2	10.2	10.5	10.8	10.1	8.8	6.8	6.3		
		MCDBR	9.8	11.8	13.9	13.9	12.7	12.1	13.1	12.8	13.0	12.1	10.1	9.0		
	5% WB	MCWBR	7.1	7.9	8.0	7.4	5.8	5.3	5.0	4.5	5.2	6.6	6.2	6.1		
		MCWBR	8.7	10.9	12.9	13.2	11.3	11.4	11.2	11.3	11.6	11.3	9.2	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.357	0.392	0.451	0.425	0.427	0.442	0.456	0.416	0.380	0.353	0.328		
	taud		2.337	2.255	2.215	2.072	2.226	2.256	2.219	2.170	2.246	2.331	2.358	2.347		
	Ebn at Noon		765	807	829	808	846	844	825	795	796	773	727	720		
	Edh at Noon		78	103	123	154	137	133	137	139	118	94	77	71		
(44) All-Sky Solar Radiation	RadAvg		1.30	2.12	3.28	4.49	5.65	6.00	6.11	5.52	4.01	2.60	1.43	1.09		
	RadStd		0.16	0.26	0.34	0.46	0.45	0.45	0.45	0.37	0.35	0.29	0.14	0.14		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.53	N/A	N/A	N/A	N/A	N/A	-114	-151	+68	+24

Nomenclature: See separate page